

exmp: prove that if $\frac{p \Rightarrow q \wedge r}{H_1}$ and $\frac{q \vee s \Rightarrow t}{H_2}$ and $\frac{p \vee s}{H_3}$ then $\frac{t}{C}$.

Proof: 1. $p \Rightarrow q \wedge r$ Hyp

2. $q \vee s \Rightarrow t$ Hyp

3. $p \vee s$ Hyp

Imp: $p \Rightarrow q \equiv \sim p \vee q$

4. $\sim(\sim p) \vee s$ Double Neg. (3)

Mp $\frac{A}{A \Rightarrow B}$
 B

5. $\sim p \Rightarrow s$ Implication (4)

6. $q \wedge r \Rightarrow q$ simplification (Tautology)

7. $p \Rightarrow q$ H.S. (1) + (6)

8. $\sim q \Rightarrow \sim p$ Contraposition (7)

9. $\sim q \Rightarrow s$ H.S. (5) + (8)

10. $q \vee s$ Implication (9)

11. t Mp (10) + (2)

Proof techniques

$$\alpha \Rightarrow \beta$$

$$\alpha \vdash \beta$$

prove that if α is T then β is T.

- direct proof $\alpha \Rightarrow \beta$ assume α is true and show β

- Indirect proof $\alpha \Rightarrow \beta \equiv \sim \beta \Rightarrow \sim \alpha$

we assume that β is false and show that α is false

- Contradiction $\alpha \Rightarrow \beta \equiv \alpha \wedge \sim \beta \Rightarrow F$

Reductio ad absurdum

assume α is T and β is F and then show a contradiction

- By cases $(\alpha_1 \vee \alpha_2 \vee \dots \vee \alpha_k) \Rightarrow \beta$

we can prove $\alpha_1 \Rightarrow \beta \wedge \alpha_2 \Rightarrow \beta \wedge \dots \wedge \alpha_k \Rightarrow \beta$

Have word: prove by truth table $(\alpha_1 \vee \alpha_2) \Rightarrow \beta \equiv (\alpha_1 \Rightarrow \beta) \wedge (\alpha_2 \Rightarrow \beta)$

α_1	α_2	β	$(\alpha_1 \vee \alpha_2) \Rightarrow \beta$	$(\alpha_1 \Rightarrow \beta) \wedge (\alpha_2 \Rightarrow \beta)$
-	-	-		
-	-	-		
-	-	-		

- Biconditional $\alpha \Leftrightarrow \beta$ (if and only if)

we prove $\alpha \Rightarrow \beta$ and we prove $\beta \Rightarrow \alpha$

$$\alpha \Rightarrow \beta \equiv \alpha \wedge \sim \beta \Rightarrow F$$

α	β	$\alpha \Rightarrow \beta$	$\alpha \wedge \sim \beta \Rightarrow F$

$$\alpha \wedge \sim \beta \Rightarrow F$$



For this statement to be correct we need F on left side

$\alpha \wedge \sim \beta$ must be false

α is an assumption and is always True

therefore $\sim \beta$ has to be false

which means β is true.

optimal

exmp: prove by contradiction that

if $\underbrace{p}_{H_1}$ and $\underbrace{p \Rightarrow q}_{H_2}$ and $\underbrace{q \Rightarrow r}_{H_3}$ then $\underbrace{r}_{\beta}$.

α

For proof by contradiction we need to show

$$\alpha \wedge \sim \beta \Rightarrow F$$

$\downarrow$

$$H_1 \wedge H_2 \wedge H_3 \wedge \sim r \Rightarrow F$$

1. p Hyp

2. $p \Rightarrow q$ Hyp

3. $q \Rightarrow r$ Hyp

4. $\sim r$ neg of conclusion

5. $\sim q$ MT (3) + (4)

6. $\sim p$ MT (2) + (5)

7. $p \wedge \sim p$ conjunction (1) + (6)

$$A \wedge \sim A \equiv F$$

8. F negation law (7)

Constructive Dilemma

$$\begin{array}{ll} p \Rightarrow q & H \\ r \Rightarrow s & H \\ p \vee r & H \\ \hline q \vee s & C \end{array}$$

Proof by contradiction:

$$1. p \Rightarrow q \quad \text{Hyp.}$$

$$2. r \Rightarrow s \quad \text{Hyp.}$$

$$3. p \vee r \quad \text{Hyp.}$$

$$4. \sim(q \vee s) \quad \text{neg. of conclusion}$$

$$5. \sim q \wedge \sim s \quad \text{D.M } \oplus$$

⋮

$$\bullet \sim(p \vee r)$$

$$\bullet (p \vee r) \wedge \sim(p \vee r) \quad \text{Conjunction } \textcircled{3}$$

$$\bullet F$$

neg. law

H.W. prove by contradiction that

if $p \Rightarrow q$ and $q \vee s \Rightarrow t$ and $p \vee s$ then t .

Proof by cases

$$(d_1 \vee d_2 \vee \dots \vee d_k) \Rightarrow \beta$$

we need to show $d_1 \Rightarrow \beta$ and $d_2 \Rightarrow \beta$ and $\dots d_k \Rightarrow \beta$

$$\text{example: } \underbrace{(p \wedge (p \Rightarrow \neg q))}_{\alpha_1} \vee \underbrace{(q \Rightarrow \neg q)}_{\alpha_2} \vdash \underbrace{\neg q}_{\beta}$$

Case 1: $\alpha_1 \vdash \beta$

$$\alpha_1 \Rightarrow \beta$$

direct proof:

$$1. p \wedge (p \Rightarrow \neg q) \text{ Hyp.}$$

$$2. p \quad \text{Simp } ①$$

$$3. p \Rightarrow \neg q \quad \text{Simp } ①$$

$$4. \neg q \quad \text{Mp } ② + ③$$

Case 2: $\alpha_2 \Rightarrow \beta$

Proof by contradiction

$$1. q \Rightarrow \neg q \quad \text{Hyp}$$

$$2. \neg(\neg q) \quad \text{neg. of conclusion}$$

$$3. q \quad \text{double neg. } ②$$

$$4. \neg q \quad \text{Mp. } ③ + ①$$

we are not done yet
we need to get to F

$$5. q \wedge \neg q \quad \text{con } ③ + ④$$

$$6. F \quad \text{neg law } ⑤$$

- exmp:
- H_1 - If Ben plays basketball, then he plays in the NBA
 - H_2 - If Ben ^Bplays hockey, then he plays in the NHL
 - H_3 - If Ben plays in the ^HNHL or NBA, then his ^Ldime is not wasted
 - H_4 - Ben's dime is wasted W

prove that: Ben plays neither basketball nor hockey. C

Direct Ben does play basketball and Ben does not play hockey

1. $B \Rightarrow N$ hyp

2. $H \Rightarrow L$ hyp

3. $L \vee N \Rightarrow \sim W$

4. W

5. $\sim(L \vee N)$ MT ③ + ④

6. $\sim L \wedge \sim N$ D.M ⑤

7. $\sim L$ Simp ⑥

8. $\sim N$ Simp ⑥

9. $\sim H$ MT. ② + ⑧

10. $\sim B$ MT. ① + ⑧

11. $\sim B \wedge \sim H$ Conj ⑨ + ⑩

• $\sim B \wedge \sim H$
 $\sim(B \vee H)$

HomeWork :

We have some evidence from a car accident :

- If the driver was conscious and was aware of the speed limit and the speedmeter was working then there was a mechanical failure.
- Inspection show that there was no mechanical failure and the speedmeter was working.

Prove that if the driver was conscious then he was not aware of the speed limit.